

Prodway AI AWS Migration Scope Definition

1. Project Overview

Migrate the Prodway platform from Railway to AWS and deploy a production-grade, scalable backend supporting multi-product APIs and agent-based workflows. Deliver a fully deployed, observable system for SowFlow and FormPilot with integrated LLM services.

2. Scope

A. Infrastructure

- Design and deploy VPC with appropriate network segmentation
- Provision compute layer using Amazon ECS
- Deploy storage layer using Amazon S3 and Amazon RDS
- Establish environment isolation across staging and production
- Configure Application Load Balancer for traffic distribution

B. Platform Migration

- Deploy containerized services from Amazon ECR to ECS
- Migrate environment variables and secrets to AWS Secrets Manager
- Configure service networking and security groups
- Execute production cutover with minimal downtime
- Validate full system functionality post-migration

C. LLM Integration

- Integrate Amazon Bedrock for LLM inference
- Implement structured request and response handling
- Optimize for latency, cost, and reliability
- Configure model access and usage monitoring

D. API Architecture

- Design scalable API structure for SowFlow and FormPilot
- Standardize request and response contracts
- Implement API versioning strategy
- Ensure extensibility for additional services
- Configure Amazon API Gateway for rate limiting and authentication

E. Agent Architecture

- Implement structured endpoints compatible with agent workflows
- Refactor services for tool-based interaction patterns
- Enable deterministic inputs and outputs
- Design idempotent operations for reliable agent execution

F. Observability

- Configure Amazon CloudWatch for logs and metrics

- Set up alerting and dashboards for system health

G. Documentation

- Generate OpenAPI specifications for all endpoints
- Produce architecture diagrams (infrastructure and application)

3. Acceptance Criteria

1. All services deployed and operational in AWS
2. LLM functionality operational via Amazon Bedrock
3. APIs documented and accessible via OpenAPI specification
4. Architecture supports agent-based workflows
5. Observability stack operational with dashboards and alerts
6. Documentation delivered and reviewed